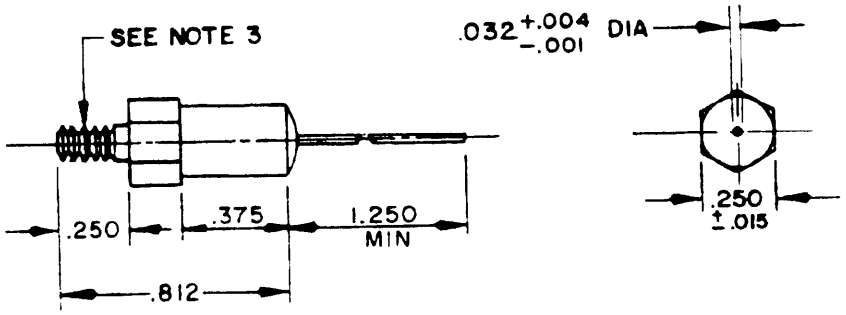
MIL-C-11015/15C
1 DECEMBER 1967
SUPERSEDING
MIL-C-11015/15B
2 April 1962

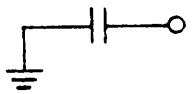
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE) (STAND-OFF), STYLE CK81

The complete requirements for procuring the capacitors described herein shall consist of this document and the issue in effect of Specification MIL-C-11015.

INCHES	MM
.001	.03
.004	.10
.015	.38
.032	.81
.250	6.35
.375	9.53
.812	20.62
1.250	31.75





CIRCUIT DIAGRAM

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Threaded portion shall be 6-32 UNC-2A and shall extend to within .047 (1.19 mm) of the shoulder.
4. A nut and internal tooth lockwasher shall be supplied with each capacitor.
5. Unless otherwise specified, tolerance is ±.031 (.79 mm).

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See above figure.
Case type - Stand-off.
Capacitance value - See table I.
Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.
Rated temperature - -55° to +85° C.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250 percent of rated voltage. applied between the case and terminal.
Body insulation: Not applicable.

Barometric pressure (reduced): Method 105 of MIL-STD-202. condition B (50, 000 ft).
Test potential - 150 percent of rated voltage, applied between the case and terminal.

FSC 5910

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TABLE I. Style CK81 characteristics.

Type designation	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pf	
CK81AX271K	500	AX	270	K
CK81AX331K	500	AX	330	K
CK81AX391K	500	AX	390	K
CK81AX471K	500	AX	470	K
CK81AX561K	500	AX	560	K
CK81AX681K	500	AX	680	K
CK81AW821M	500	AW	820	M
CK81AW102M	500	AW	1,000	M

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 20,000 megohms, min. measured between the insulated point and capacitor case.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency - 1 kHz ± 100 Hz.
Test voltage - 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 1.5 percent, max (voltage-temp limit X) or 2.0 percent, max (voltage-temp limit W).

Shock, medium impact: Method 205 of MIL-STD-202, condition C (50 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D.
Immersion: Method 104 of MIL-STD-202, condition B.
IR - 3,000 megohms, min.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV - 250 percent of rated voltage.
IR - 3,000 megohms, min.
Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202. 1 terminal.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
IR - 20,000 megohms, min.
ΔCap. - ±5 percent of initial measurement.
ΔDF - 0.5 percent, max.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).
Test potential - 200 percent of rated voltage.
IR - 500 megohms, min (at 85° C) and 10,000 megohms, min (at 25° C).
DF - 3.0 percent, max (voltage-temp limit X) or 4.0 percent, max (voltage-temp limit W) (at 25° C).

Marking: In accordance with MIL-C-11015.

NOTE: Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MI
Navy - EC, AS, OS, SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

(Project 5910-0829-13)

Code "C"

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